JASPER ANDRA

AI/ML ENGINEER | GENAI • RAG • AGENT SYSTEMS

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SUMMARY

AI/ML Engineer building production-ready generative AI systems with Python, retrieval-augmented generation (RAG), large language models (LLMs), and multi-agent workflows. Combined end-to-end Python application ownership with security-aware design, guardrails, and MLOps. Prepared to deliver reliable AI products in distributed, remote engineering teams.

TECHNICAL SKILLS

GenAI & Agents: RAG, LangChain, CrewAI, multi-agent workflows, context engineering, LoRA/QLoRA, LLM guardrails, ChromaDB
ML & Data: PyTorch, TensorFlow, scikit-learn, BERT, ensemble methods, neural collaborative filtering, Pandas, NumPy, ETL
Backend: Python, FastAPI, Flask, SQL, PostgreSQL, Bash
MLOps & Cloud: Docker, MLflow, Airflow, GitHub Actions, CI/CD, GCP Cloud Build, AWS SageMaker
Security: IDOR/BFLA protection, PII isolation, OTP verification (Twilio/Elastic Email), SOPS secrets management

EXPERIENCE

Research Assistant | Webster University | 05/2024 – 05/2026

- Conducted applied research on machine learning for threat detection, anomaly identification, and security automation.
- Prototyped and benchmarked models for vulnerability discovery and threat classification under real-world evaluation conditions.
- Built a reusable Python framework for data analysis and model validation, improving experiment consistency and repeatability.
- Presented findings and evaluation methods to faculty and stakeholders, translating technical results into actionable conclusions.

Python Developer | Concentrix | 05/2022 – 06/2024

- Owned Python applications from requirements through deployment, automating core business workflows and reducing manual effort.
- Designed Flask REST APIs backed by SQL databases, enabling internal teams to access and integrate operational data.
- Built Pandas and NumPy pipelines that transformed raw inputs into validated reports for business use.
- Improved production reliability through structured error handling, logging, automated tests, and Agile incident diagnosis.

PROJECTS

Secure RAG Knowledge Assistant | *Python, ChromaDB, sentence-transformers, FastAPI, Docker*

- Built a working RAG pipeline: overlapping chunking of .txt/.md/.pdf documents, local all-MiniLM-L6-v2 embeddings, and a persistent ChromaDB vector store with idempotent ingestion.
- Implemented custom input/output guardrails: prompt-injection and jailbreak blocking, PII redaction (email, phone, SSN, credit card) before LLM access, and off-topic refusal via embedding distance.
- Exposed /ask (answers with citations) and /health through FastAPI, with a pluggable OpenAI provider interface and Docker deployment.
- Wrote 22 offline pytest tests covering chunking, ingestion, retrieval, guardrails, and API endpoints.

Multi-Agent Workflow Automation | *CrewAI, Python, PostgreSQL, GitHub Actions*

- Orchestrated triage, research-specialist, and reviewer agents with tool calling, retry-once-then-fallback logic, and confidence-based review flags.
- Built a deterministic offline stub mode (zero network, standard library only) alongside the CrewAI/OpenAI backend for end-to-end runs without API keys.
- Persisted ordered, foreign-key-linked audit trails to PostgreSQL via SQLAlchemy (SQLite fallback for local development), with human approval before execution.
- Wrote 16 pytest tests covering agent routing, persistence round-trips, approval logging, and run determinism.

ML Threat Detection Benchmark | *PyTorch, scikit-learn, MLflow, Pandas, NumPy*

- Built a reproducible benchmark comparing a RandomForest ensemble, PyTorch MLP, and IsolationForest on seeded synthetic network telemetry with ~11% threats.
- Implemented ETL with deduplication, imputation, feature engineering, and stratified 70/15/15 Parquet splits; logged run parameters, metrics, and artifacts in MLflow.
- Evaluated per-class precision, recall, F1, ROC-AUC, and confusion matrices; surfaced the anomaly detector ranking-versus-classification trade-off.
- Wrote 13 pytest tests covering ETL determinism, training, evaluation, and MLflow logging.

EDUCATION

M.S. Cybersecurity, Emphasis in Artificial Intelligence | 08/2023 – 08/2026
Webster University | St. Louis
Coursework: Machine Learning, Deep Learning, Artificial Intelligence with Python, Cybersecurity Threat Detection, Databases and Data Warehouses

B.Tech, Computer Science Engineering | 05/2019 – 05/2023
Vignan's Nirula Institute of Technology | India
Coursework: Database Systems, Algorithm Design, Computer Networks, Machine Learning, Object-Oriented Programming (Java)